Hardware Documentation

PARMCO Project — Checkpoint 2
AI-Generated Documentation
Subsystem: Motor, Isolation & IR Sensor

Table of Contents

1. System Overview 1
2. Bill of Materials 1
3. GPIO Assignment 2
4. Pin-to-Pin Assembly Guide 2
5. Assembly and Bring-Up Instructions 3
6. How It Works (Functional Analysis) 4
6.1 Power and isolation logic (Quiet Start) 4
6.2 Bipolar drive sequence 4
6.3 Speed and step timing 5
6.4 Current, power and decoupling 5
6.5 IR sensor and RPM measurement 5
Schematic: Isolation, Drive & IR Sensor 6

1. System Overview

The PARMCO hardware architecture safely interfaces a 3.3V Raspberry Pi 4 with a 12V Nema17 bipolar stepper motor (XY42STH34-0354A: 1.8° per step, rated 12 V and 0.35 A per phase). A 4N25 optocoupler is the on/off enable to the H-bridge: the Pi's GPIO enable signal only drives the optocoupler LED, so the motor stays disabled during the Pi's boot sequence ("Quiet Start"). An SN754410NE Quad Half-H Bridge translates the logic-level half-step sequence from four GPIO pins into the high-current phases that drive the two motor coils.

Checkpoint 2 adds an **IR sensor** that detects the two propeller blades and feeds one digital pulse per blade to a Pi GPIO input, so the software can report propeller RPM. The phone link uses the Pi's on-board Bluetooth radio and needs no additional circuitry. The AS5600 magnetic encoder is added at Checkpoint 3 and is not part of this circuit.

Changes from Checkpoint 1: IR sensor added (parts list, GPIO assignment, pin table, schematic); the enable pull-down resistor and the supply decoupling capacitors are documented; the half-step table matches the Checkpoint 2 code; GPIO numbers now match the Checkpoint 2 code; assembly and bring-up instructions added; the "how it works" analysis is expanded, including a corrected description of what the 4N25 isolates (section 6).

2. Bill of Materials

Component	Purpose	Specification
Raspberry Pi 4	Main processing unit. Runs the C/ARM assembly control program, reads the IR sensor, hosts the Bluetooth server.	3.3V Logic via Protector HAT; on-board Bluetooth
Nema17 Stepper Motor	Primary actuator, modified with 3D-printed stand.	XY42STH34-0354A, 1.8° Step (Bipolar), 12 V, 0.35 A per phase
SN754410NE	Quad Half-H Bridge. Drives current to motor coils (two H-bridges).	Min 1A Output Capacity; VCC1 logic 5 V, VCC2 motor 12 V; internal output clamp diodes

Component	Purpose	Specification
4N25 Optocoupler	Enable path to the H-bridge; keeps the motor disabled at boot.	NPN Phototransistor, 6-pin DIP (pin 3 NC, pin 6 base left unconnected)
Resistor R1	Limits GPIO17 current through the 4N25 LED.	320 ohm (about 6.6 mA LED current at 3.3 V)
Resistor R2	Pull-down on the 4N25 emitter / SN754410NE enable node (pins 1 and 9): holds the enables low whenever the 4N25 is off.	10 kohm, 4N25 pin 4 to ground
Capacitor C1	Decoupling for the 12V motor rail (VCC2).	100 µF, rated above 12 V, polarized (+ to 12V)
Capacitor C2	Decoupling for the 5V logic rail (VCC1).	0.1 µF
IR Sensor module	Detects propeller blades: 2 pulses per revolution, used for RPM.	Digital output, HIGH while a blade is detected; powered from the Pi 3.3V pin
DC Power Supply	Provides dual-rail power to the logic and motor circuits.	CH1: 5V (0.2A), CH2: 12V (1A)
2-Blade Propeller	Rotational visual indicator (trimmed for clearance); one blade marked as the reference blade.	Plastic, Friction Fit
Breadboard and jumper wires	Common ground rail and circuit interconnect.	Solderless breadboard
Android phone	Bluetooth RFCOMM client (not part of the circuit).	Connects to the Pi's on-board Bluetooth; no extra wiring

3. GPIO Assignment

GPIO numbers are the BCM numbers used by the class RP4 library and by the Checkpoint 2 code.

Signal	BCM GPIO	Header Pin	Direction	Connects to
Motor enable	GPIO17	11	Output	4N25 Pin 1 (Anode) via R1 (320 ohm)
Coil input 1A	GPIO22	15	Output	SN754410NE Pin 2
Coil input 2A	GPIO23	16	Output	SN754410NE Pin 7
Coil input 3A	GPIO24	18	Output	SN754410NE Pin 10
Coil input 4A	GPIO25	22	Output	SN754410NE Pin 15
IR sensor pulse	GPIO27	13	Input	IR Sensor OUT
3.3V supply	—	1	Power	IR Sensor VCC
Ground	—	6	Ground	Common Ground Rail

4. Pin-to-Pin Assembly Guide

Source Component	Source Pin / Rail	Destination Component	Destination Pin
DC Power Supply	12V (VCC2)	SN754410NE	Pin 8 (VCC2 - Motor Power)
DC Power Supply	5.0V (VCC1)	SN754410NE	Pin 16 (VCC1 - Logic Power)
DC Power Supply	Ground (-)	Breadboard Rail	Common Ground
Raspberry Pi 4	Ground (Pin 6)	Breadboard Rail	Common Ground

Source Component	Source Pin / Rail	Destination Component	Destination Pin
Raspberry Pi 4	GPIO17 Output (Enable), Pin 11	4N25 Optocoupler	Pin 1 (Anode) via R1 (320 ohm)
4N25 Optocoupler	Pin 2 (Cathode)	Breadboard Rail	Common Ground
4N25 Optocoupler	Pin 5 (Collector)	DC Power Supply	5.0V (VCC1)
4N25 Optocoupler	Pin 4 (Emitter)	SN754410NE	Pins 1 & 9 (Enables)
Raspberry Pi 4	GPIO22, 23, 24, 25 (Step Seq)	SN754410NE	Pins 2, 7, 10, 15 (1A, 2A, 3A, 4A) respectively
SN754410NE	Pins 1 & 9 (enable node, 4N25 Pin 4)	Resistor R2 (10 kohm)	Breadboard Rail (Common Ground)
SN754410NE	Pins 4, 5, 12, 13	Breadboard Rail	Common Ground (Heatsink)
DC Power Supply	12V (VCC2) rail	Capacitor C1 (100 µF)	+ to 12V, - to Common Ground
DC Power Supply	5.0V (VCC1) rail	Capacitor C2 (0.1 µF)	One lead to 5V, one lead to Common Ground
SN754410NE	Pins 3 & 6 (1Y, 2Y)	Nema17 Motor	Coil A Pair
SN754410NE	Pins 11 & 14 (3Y, 4Y)	Nema17 Motor	Coil B Pair
Raspberry Pi 4	3.3V (Pin 1)	IR Sensor	VCC
IR Sensor	GND	Breadboard Rail	Common Ground
IR Sensor	OUT	Raspberry Pi 4	GPIO27 (Pin 13), input

5. Assembly and Bring-Up Instructions

Wear eye protection whenever a propeller is spinning. Fit the propeller only after the quiet start and the stop command have been confirmed. Keep the DC supply outputs off and the RP4 unplugged until step 10.

Build (all power off)

1. Attach the Nema17 to the 3D-printed stand. Identify the two coil pairs by continuity with a meter (a coil pair reads a low resistance; wires from different pairs read open).
2. Place the SN754410NE and 4N25 on the breadboard, observing the pin-1 notch/dot of each. Connect the breadboard ground rail to the DC supply ground (-), RP4 Pin 6, and SN754410NE pins 4, 5, 12 and 13.
3. Connect supply CH2 (12V) to SN754410NE pin 8 and CH1 (5V) to pin 16. Neither supply pin may connect to any RP4 pin.
4. Place decoupling capacitor C1 (100 µF, positive lead to the 12V rail) between the 12V rail and the ground rail, and C2 (0.1 µF) between the 5V rail and the ground rail, close to SN754410NE pins 8 and 16.
5. Wire the 4N25: RP4 GPIO17 (Pin 11) through resistor R1 (320 ohm) to pin 1; pin 2 to ground; pin 5 to the 5V (VCC1) rail; pin 4 to SN754410NE pins 1 and 9. Connect pull-down resistor R2 (10 kohm) from pin 4 to the ground rail. Leave 4N25 pins 3 and 6 unconnected.
6. Wire GPIO22, 23, 24, 25 (Pins 15, 16, 18, 22) to SN754410NE pins 2, 7, 10, 15.
7. Wire coil A to SN754410NE pins 3 and 6, and coil B to pins 11 and 14.
8. Wire the IR sensor: VCC to RP4 Pin 1 (3.3V), GND to the ground rail, OUT to GPIO27 (Pin 13). Position the sensor so each blade passes through its detection zone (two pulses per revolution).
9. Pre-power check: confirm ground continuity (RP4, supply, pins 4/5/12/13); confirm VCC1 and VCC2 trace only to the DC supply; confirm no 5V or 12V wire touches any RP4 pin; confirm the polarity of C1 and that R2 is in place;

with a meter check that the 3.3V/GND/OUT wires of the IR sensor are not swapped.

Power-up and verification

10. Set CH1 to 5.0 V and CH2 to 12.0 V (current limits as the instructor directs), then turn the outputs on with the RP4 still unplugged. VCC2 current must be below 0.05 A and the shaft must turn freely by hand. If the shaft locks or the current is 0.5 A or more, the enable wiring is wrong: power down.
11. Plug in the RP4 and watch the shaft and VCC2 current for the whole boot: no movement, current below 0.05 A, shaft free afterwards (Quiet Start).
12. Start the control program. Enable the motor: VCC2 current rises (roughly 0.3 A with one coil energized and 0.6-0.7 A with two, depending on the active phase) and the shaft holds. Disable: current returns below 0.05 A.
13. Run one revolution each way and compare the stop position with the start mark (one revolution is 400 half steps). Then run continuously at a moderate rate and compare the IR RPM on the terminal with the commanded RPM (steps per second x 0.15, see section 6.3).
14. Power down: disable the motor, shut down the RP4, then turn off the DC supply outputs.

6. How It Works (Functional Analysis)

6.1 Power and isolation logic (Quiet Start)

The SN754410NE drives current only while its enable inputs (pins 1 and 9) are high. Those inputs are fed by the emitter of the 4N25 phototransistor, whose collector is tied to the 5V logic rail (VCC1), and they are held to ground by the 10 kohm pull-down resistor R2. The RP4 GPIO17 pin drives only the 4N25 LED, through R1 (320 ohm) to ground. During Pi boot the GPIO pins float or are pulled low, so the LED is off and the phototransistor is open. R2 then holds the enable inputs at ground, far below the logic-low threshold, so the H-bridge outputs stay off and zero current is delivered to the motor coils, regardless of the 12V VCC2 state or the state of the four coil inputs. When the program sets GPIO17 high, about 6.6 mA ((3.3 V - 1.2 V) / 320 ohm) flows in the LED and the phototransistor conducts. Its photocurrent (about 1 mA or more at the 4N25's minimum 20% transfer ratio) exceeds the 0.5 mA that R2 draws at 5 V, so the enable inputs rise to roughly 5 V minus the transistor's saturation voltage, well above the logic-high threshold of the SN754410NE.

The 4N25 gives a signal path in which the RP4 pin never connects to the 5V or 12V rails, and the enable is on/off only (it is never PWM'd). The grounds of the RP4, the supply and the H-bridge are common (required, because the 3.3V step inputs need a shared reference), so this is optical coupling of the enable signal, not full galvanic isolation of the two grounds.

6.2 Bipolar drive sequence

Each coil of the bipolar Nema17 is connected between the two outputs of one H-bridge: coil A between 1Y and 2Y (pins 3 and 6), coil B between 3Y and 4Y (pins 11 and 14). An input driven high pulls its output to VCC2, an input driven low pulls it to ground. For coil A: 1A high and 2A low gives current in the + direction (A+); 1A low and 2A high gives A-; both inputs equal gives no current. Coil B behaves the same with 3A and 4A. The program steps through the 8-entry half-step table below, one entry per half step. Bit 3 of each table byte drives 1A (GPIO22), bit 2 drives 2A (GPIO23), bit 1 drives 3A (GPIO24) and bit 0 drives 4A (GPIO25). The table index advances by one for one direction and goes back by one for the other, wrapping from 7 to 0 and 0 to 7. One table pass is 8 half steps; 400 half steps (0.9° each) make one revolution.

Index	1A 2A 3A 4A	Coil A	Coil B	Byte
0	1 0 0 0	A+	off	0x08
1	1 1 0 0	off	off	0x0C
2	0 1 0 0	A-	off	0x04

Index	1A 2A 3A 4A	Coil A	Coil B	Byte
3	0 1 1 0	A-	B+	0x06
4	0 0 1 0	off	B+	0x02
5	0 0 1 1	off	off	0x03
6	0 0 0 1	off	B-	0x01
7	1 0 0 1	A+	B-	0x09

```
// Bit mapping: [3]1A, [2]2A, [1]3A, [0]4A
// Sequence: 1000, 1100, 0100, 0110, 0010, 0011, 0001, 1001
half_step_table: .byte 0x08, 0x0C, 0x04, 0x06, 0x02, 0x03, 0x01, 0x09
```

6.3 Speed and step timing

In half-step mode, RPM = steps per second x 60 / 400 = steps per second x 0.15. The program's default of 133.3 steps/s is 20 RPM; 100 RPM is 666.7 steps/s, or 1.5 ms between steps.

6.4 Current, power and decoupling

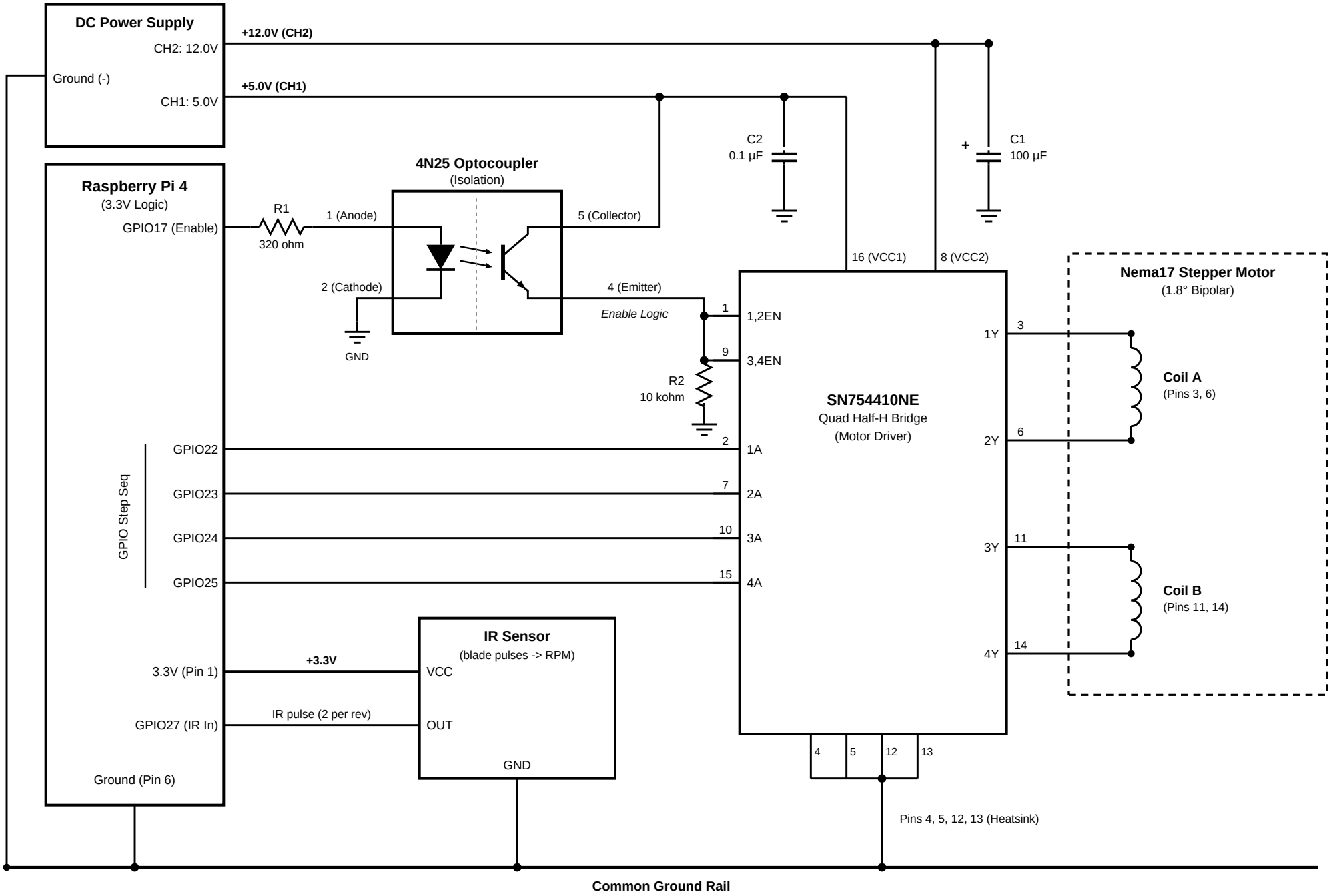
At rated 12 V and 0.35 A per phase, a single energized coil draws about 0.3-0.35 A (less the H-bridge saturation drop) and a two-coil phase about 0.6-0.7 A. This is within the 1 A supply channel and the 1 A per-channel rating of the SN754410NE. VCC1 (5V) supplies the SN754410NE logic and the 4N25 collector current, far below the 0.2 A limit. GPIO17 sources about 6.6 mA to the LED, within the RP4 per-pin limits. The SN754410NE has internal output clamp diodes for the inductive flyback of the coils; ground pins 4, 5, 12 and 13 on the common rail also conduct heat away from the package. Check the package for heat during long runs. Capacitor C1 (100 µF) across the 12V rail supplies the brief surges of current drawn when the coils switch and keeps VCC2 steady against the voltage spikes that coil switching causes; capacitor C2 (0.1 µF) across the 5V rail bypasses high-frequency switching noise at the SN754410NE logic supply pin.

6.5 IR sensor and RPM measurement

The IR sensor is powered from the Pi's 3.3V pin so its output swings 0 to 3.3 V and can be connected directly to GPIO27 (the Pi inputs are not 5V tolerant). Each time a blade passes through the sensor's detection zone the output produces a pulse; the software polls GPIO27 every 1 ms and treats a low-to-high change as one blade. Because the propeller has two blades, consecutive pulses are half a revolution apart, so RPM = 30 / (seconds between pulses). The pulse spacing is 0.3 s at 100 RPM and 1.5 s at 20 RPM, so a 1 ms poll is ample. The sensor output is HIGH while a blade is in the detection zone, which is the polarity the software expects, and the sensor ground joins the common ground rail.

PARMCO Hardware Isolation, Drive & IR Sensor Schematic

Subsystem: Motor, Isolation & IR Sensor (Checkpoint 2)



GPIO numbers are BCM numbers as used by the class RP4 library. The AS5600 encoder is not wired at Checkpoint 2 (added at Checkpoint 3).